

publish-subscribe methodology. DDS can be deployed in platforms ranging from low-footprint devices to the cloud, and supports efficient bandwidth usage as well as the agile orchestration of system components.

Other open source IoT communication protocols are VSCP, STOMP, etc.

Open source infrastructure

The following are some of the key open source cloud platforms.

OpenStack: This is open source software for creating private and public clouds, built and disseminated by a large and democratic community of developers, in collaboration with users. OpenStack is mostly deployed as Infrastructure-as-a-Service (IaaS), whereby virtual servers and other resources are made available to customers.

Cloud Foundry: This is an open Platform-as-a-Service (PaaS), which provides a choice of clouds, developer frameworks and application services. Cloud Foundry makes it faster and easier to build, test, deploy and scale applications.

OpenShift: This is Red Hat's cloud computing PaaS offering. It is an application platform in the cloud, where app developers and teams can build, test, deploy and run their applications.

Cloudify: Cloudify was developed and designed on the principles of openness to power the IT transformation revolution. It enables organisations to design, build and deliver various business applications and network services.

Other open source cloud platforms are WSO2, Cloud Stack, etc.

Open source hardware

Open source hardware consists of physical artefacts of technology designed and offered by the open design movement.

Arduino: This operates through an array of hardware specifications that can be given to interactive electronics. The software comes in the form of the Arduino programming language and the Arduino Integrated Development Environment (IDE).

Raspberry Pi: This is a very low cost computer that runs Linux, but it also provides a set of GPIO (general purpose input/output) pins that allow you to control electronic components for physical computing and explore the Internet of Things (IoT).

mangOH: This industrial grade open source hardware is designed to address common IoT pain points and deliver out-of-the-box IoT prototypes of applications, to bring IoT products to market sooner.

Open source operating systems

The following are the open source operating systems.

Android Things (formerly known as Brillo): This is an operating system from Google built on the power of Android for embedded devices. It has access to all the Android APIs, Google services and developer tools.

RIOT (Real-time IoT): This is a real-time operating system, which is based on micro-kernel architecture

for IoT. RIOT is developed by a consortium of universities in Germany and France.

Zephyr: This is another real-time operating system (RTOS) similar to RIOT, launched by the Linux Foundation and available under the Apache licence. It is small, scalable and optimised for resource constrained devices.

Ubuntu Core: This is based on a Linux OS developed for IoT devices. It is delivered as Linux application packages called Snaps.

Open source is well-established in the IoT space. There are many opportunities for enterprises to use open source IoT to make business services more efficient and customer-centric. Open source IoT can help enterprises achieve their most important goals, which are:

- Better customer experience, a unified product and service experience
- Huge cost savings
- Improving customer requests and service delivery
- Improving innovation
- The capability to deliver and manage highly secure, reliable, and scalable IoT solutions
- Gain efficiency and stimulate innovation 

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